MIL-I-22668A (WEP)
15 MARCH 1963
SUPERSEDING
MIL-I-22668(WEP)
20 OCTOBER 1960

MILITARY SPECIFICATION

INTEGRATED ELECTRONIC CENTRALS

AN/ASQ-56(*), AN/ASQ-57(*) and AN/ASQ-58(*)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The Electronic Centrals covered by this specification shall provide communication, navigation, and identification facilities required in naval aircraft to carry out military missions.

1.2 Classification - The Integrated Electronic Centrals shall be of three types which together shall include the following equipments:

Item	Name	Type Designation
1	Receiver-Transmitter, Radio	RT-542/ASQ
2	Receiver-Transmitter, Radio	RT-541/ASQ
3	Decoder, Pulse	KY-309/ASQ
4	Coder-Receiver-Transmitter	KY-308(*)/ASQ
5	Amplifier-Power Supply-Receiver	AM-2310/ASQ
6	Antenna	AS-909/ARA-48
7	Control, Radio Set	C-3146/ASQ
8	Mounting-Plenum	MT-2243/ASQ-56
9	Control, Radio Set	C-2790/ARC
10	Switching Unit	SA-629/ARC
11	Mounting	MT-2141/ARC
12	Indicator, Frequency Channel	ID-758/ARC
13	Mounting-Plenum	MT-2244/ASQ

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Item	Name	Type Designation
14	Control, Radio Set	C-2791/ARC
15	Receiver-Transmitter, Radio	RT-559/ASQ-58
16	Control, Radio Set	C-1607/ARC-52
17	Control, Radar Set	C-1159/APX-6B
18	Control, Radar Set	C-1272A/APA-89
19	Control, Receiver	C-1457/ARR-40

1.3 Types - The Electronic Centrals covered by this specification shall include the following equipments. Each central shall consist of the items and quantities as specified below:

*Items Required for Each Central

AN/ASQ-56	AN/ASQ-57	AN/ASQ-58
Item 1	Item 1	Item 1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7 (2)	7	7
8	9	13
9 (2)	10	14
10	11	15
11	12	16
12 (2)	13	17
17	17	18
18	18	
19 (2)	19	

*Item numbers correspond to items of paragraph 1.2. Quantity of each item is one unless otherwise designated by quantity in parenthesis.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein: (See 6.5(4))

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SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-M-7793	Meter, Time Totalizing
MIL-R-7893	Radar Identification Set AN/APX-6B
MIL-C-7953	Coder Group AN/APA-89
MIL-P-17555	Preparation for Delivery of Electronic, Electrical, and Electro-Mechanical Equipment and Associated Repair Parts
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-R-18400	Receiving Set, Radio AN/ARR-40
MIL-R-18401	Radio Set AN/ARC-52
MIL-E-19600	Electronic Modules, General Specification For
MIL-A-22669	Amplifier-Power-Supply-Receiver AM-2310/ASQ
MIL-C-22670	Control, Radio Set C-2790/ARC; Indicator, Frequency Channel ID-758/ARC; Mounting MT-2141/ARC and Switching Unit SA-629/ARC

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MIL-C-22671	Control, Radio Set C-3146/ASQ
MIL-C-22672	Coder-Receiver-Transmitter KY-308(*)/ASQ
MIL-M-22673	Mounting-Plenum MT-2243/ASQ-56
MIL-M-22674	Mounting-Plenum MT-2244/ASQ
MIL-R-22677	Receiver-Transmitter, Radio RT-541/ASQ and Decoder, Pulse KY-309/ASQ
MIL-R-22678	Receiver-Transmitter, Radio RT-542/ASQ
MIL-R-22679	Receiver-Transmitter, Radio RT-559/ASQ-58
MIL-E-23094	Reliability Assurance for Production Acceptance of Avionic Equipment

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-704	Electronic Power, Aircraft, Characteristics and Utilization of
MS-25245	Connections, Input Power Airborne Electronic Equipment; 28-volt DC, 115-volt Single Phase AC, or 115/200-volt Three-Phase AC, or combinations
MS-17322 ( )	Meter, Time Totalizing Miniature 400 Cycle, Digital

2.2 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code DCS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and material requirements shall conform to Specification MIL-E-5400.

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(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to-failure of more than 5000 hours should be the goal of each.

(3) When previously produced models of these equipments did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction -

3.2.1 General - The centrals shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction, and workmanship unless otherwise specified herein or in the individual equipment specifications.

*3.2.2 Total Weights - The total weight of each central, excluding cables, shall be a minimum consistent with good design and shall not exceed 250.0 pounds for the AN/ASQ-56, 250.0 pounds for the AN/ASQ-57 and 284.4 pounds for the AN/ASQ-58.

*3.2.3 Modular Construction - The use of modular space assignment and construction of easily disconnected plug-in subunits or component assemblies shall be in accord with Specification MIL-E-19600 where possible. The following considerations shall apply where practicable; except that for the KY-308(*)/ASQ, ease of maintenance shall be the primary consideration:

(1) Modules shall be of standardized form and dimensions to facilitate use in other equipment of different over-all configuration and shall be designed for maximum practical effectiveness of cooling air supplied by or to the equipment.

(2) Each module shall contain the circuitry of one or more functions of the equipment to facilitate testing and repair. The equipment shall be divided into as many modular units as is electrically and physically practical in keeping with efficient space utilization, equipment performance and reliability.

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(3) It shall be possible to operate the equipment with one or more modules removed from the chassis but connected to it electrically by means of one or more extension cable assemblies and by use of mechanical extenders when required.

(4) It shall be possible to replace any module with other modules of the same type and obtain specified equipment performance with only minor readjustments.

3.2.4 Operating Life -

3.2.4.1 Operational Stability - The centrals shall operate with optimum performance for 50 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.4.2 Total Operating Life - The centrals shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

*3.2.4.3 Time Totalizing Meter - A time totalizing meter conforming, as applicable, with Specification MIL-M-7793 and the indicated MS number shall be included in the following:

RT-542/ASQ	MS17322
RT-541/ASQ	MS17322
KY-308(*)/ASQ	MS17322
KY-309/ASQ	MS17322
AM-2310/ASQ	MS17322
RT-559/ASQ-58	MS17322

*3.2.4.4 Reliability in Mean Time Between Failures - The centrals shall have a mean (operating) time between failure when tested and accepted as outlined under the requirements of 4.4.3 of 75 hours for the ASQ-56, 75 hours for the ASQ-57 and 60 hours for the ASQ-58.

3.2.5 Cabling and Connections -

3.2.5.1 Cables and Connectors - The centrals shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400 unless otherwise specified in the individual equipment specifications.

3.2.5.2 Interconnection Cabling - The centrals shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External

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cables and that portion of the connectors attached to the cables shall not be supplied as part of the centrals.

3.2.5.3 Input Power Connection - Input power connections shall be in accordance with MS-25245.

3.2.6 Soldering (Electrical Connections) - Secure mechanical connections shall not be required before soldering small lightweight parts to printed wiring assemblies provided equipment performance is not compromised.

3.2.7 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

*3.2.8 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181B unless otherwise specified in the individual equipment specifications.

3.2.9 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.10 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307. (See 6.8)

*3.2.11 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

*3.2.12 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient (30 ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 ± 1.0 v, 380-420 cps, 3Ø 26 ± 1.0 v, 380-420 cps, 1Ø 27.5 ± 0.5 v DC
System Cooling Air	110 lbs. per hour

*3.2.13 Service Conditions - The centrals shall operate satisfactorily under any of the environmental conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for the Classes of Equipment as specified herein unless otherwise specified in the individual equipment specifications:

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Receiver-Transmitter, Radio	RT-542/ASQ	Class 2
Receiver-Transmitter, Radio	RT-541/ASQ	Class 2
Decoder, Pulse	KY-309/ASQ	Class 2
Coder-Receiver-Transmitter	KY-308(*)/ASQ	Class 2
Amplifier-Power-Supply-Receiver	AM-2310/ASQ	Class 2
Antenna	AS-909/ARA-48	Class 2
Control, Radio Set	C-1607/ARC	Class 1
Mounting-Plenum	MT-2243/ASQ-56	Class 2
Control, Radio Set	C-2790/ARC	Class 1
Switching Unit	SA-629/ARC	Class 2
Mounting	MT-2141/ARC	Class 2
Indicator, Frequency Channel	ID-758/ARC	Class 1
Control, Radio Set	C-3146/ASQ	Class 1
Mounting-Plenum	MT-2244/ASQ	Class 2
Control, Radio Set	C-2791/ARC	Class 1

*3.2.13.1 System Cooling Air Requirements - Cooling air requirements for the AN/ASQ-56, the AN/ASQ-57 and AN/ASQ-58 are given in Figures 1, 2 and 3 of this specification.

3.2.13.2 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL-E-5400.

3.2.14 Warm-up Time - The time required for the centrals to warm up prior to operations shall be kept to a minimum and shall not exceed 5 minutes under standard conditions and 15 minutes at extreme service conditions.

*3.2.15 Primary Input Power Requirements - The centrals shall meet all applicable requirements of MIL-STD-704 unless otherwise specified herein and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required by each central shall not exceed the specified amounts.

(1) AC Power (Three Phase) 115/200 V, Category "B"

Power (3Ø)	AN/ASQ-56	900 V.A.
	AN/ASQ-57	900 V.A.
	AN/ASQ-58	1075 V.A.

(2) AC Power (Single Phase) 26 V, within limits proportional to Category "B"

*Power (1Ø)	AN/ASQ-56	100 V.A.
	AN/ASQ-57	100 V.A.
	AN/ASQ-58	120 V.A.

*Includes typical system instrumentation.

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(3) DC Power 28 V, Category "B" (see 6.2)

**Current, nominal	AN/ASQ-56	4.5 amps
	AN/ASQ-57	4.5 amps
	AN/ASQ-58	5.0 amps

**While channeling the required current per equipment shall not exceed 8 amperes

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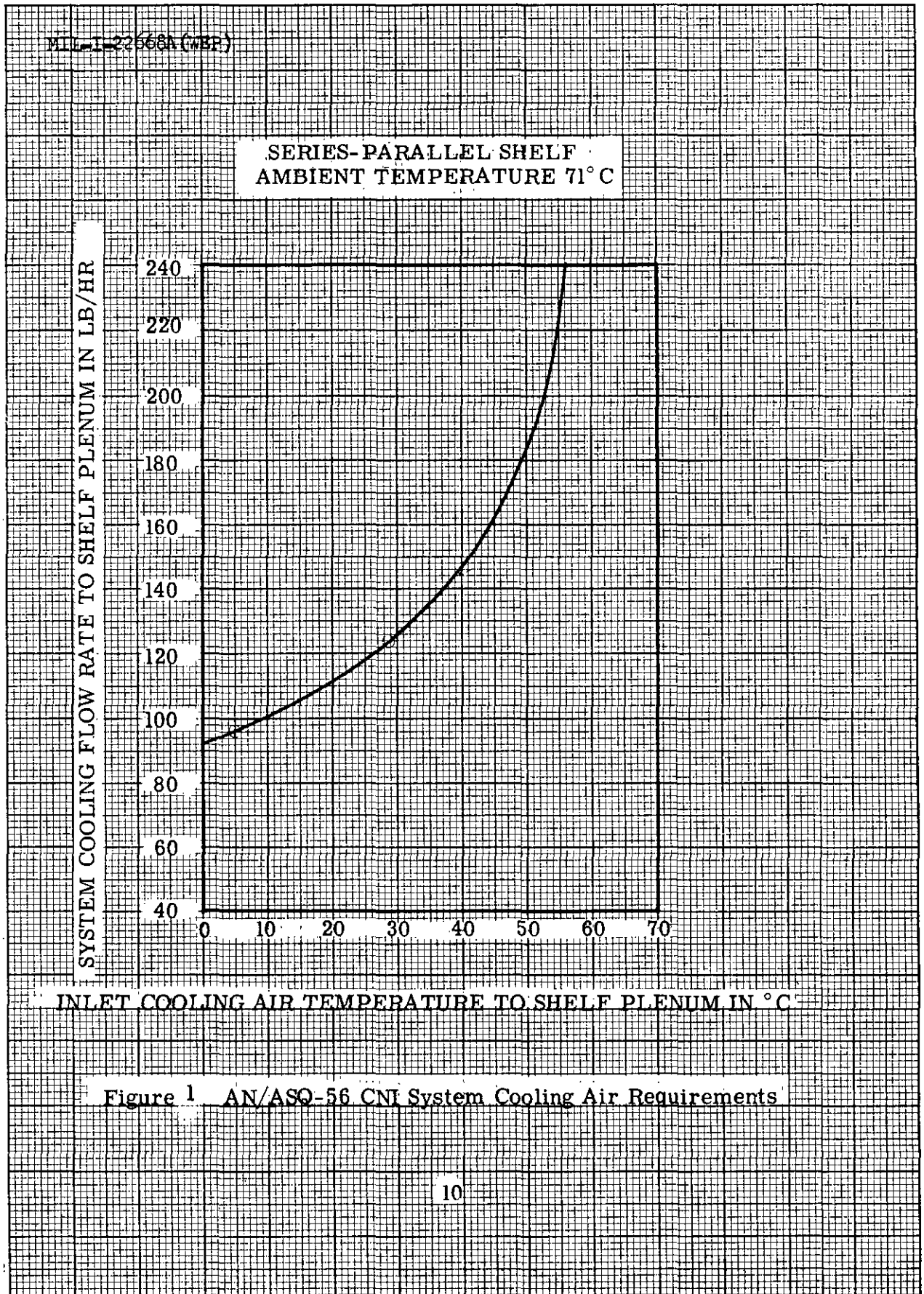
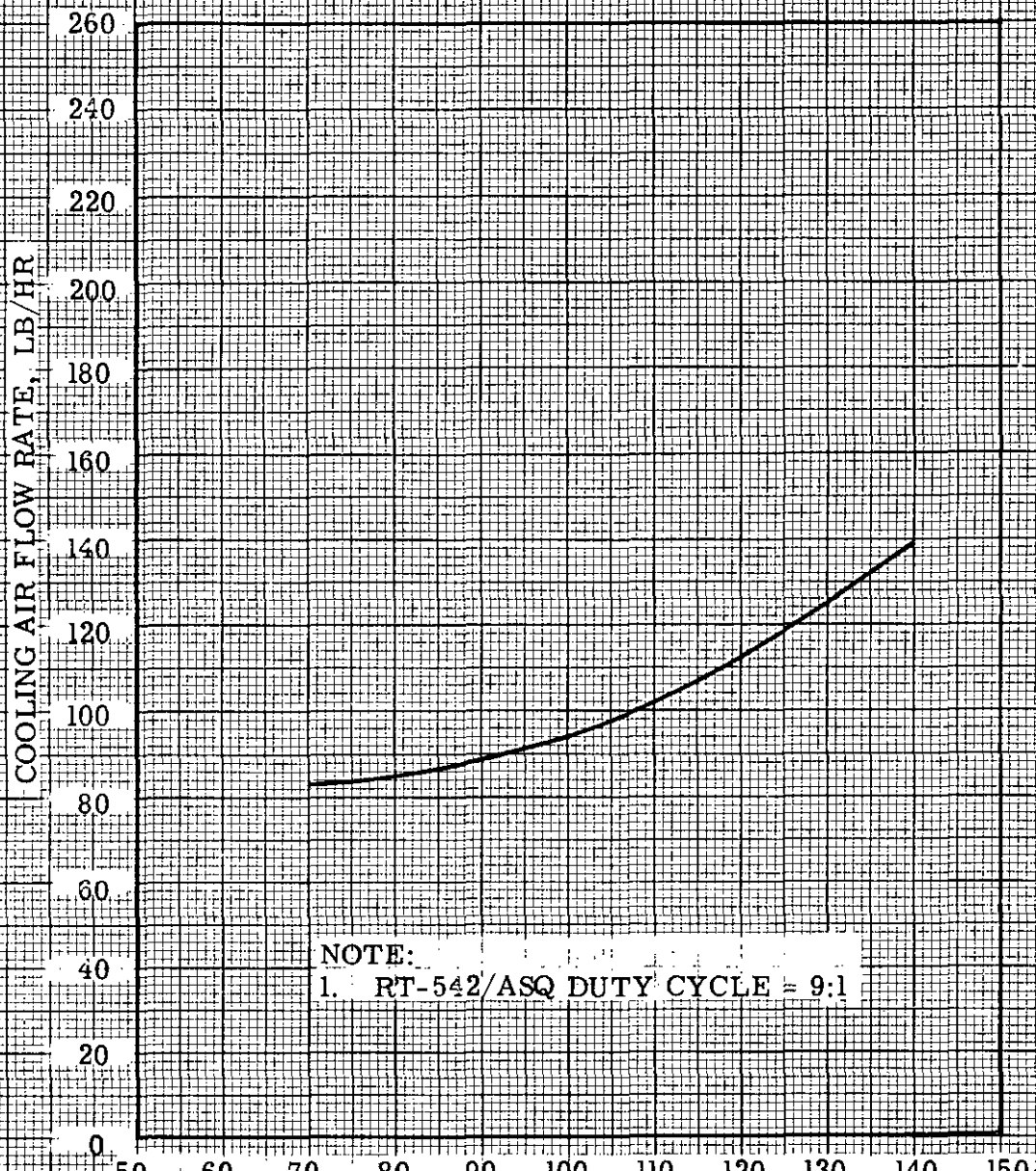


Figure 1 AN/ASQ-56 CNI System Cooling Air Requirements

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AN/ASQ-57 CNI SYSTEM COOLING AIR REQUIREMENTS

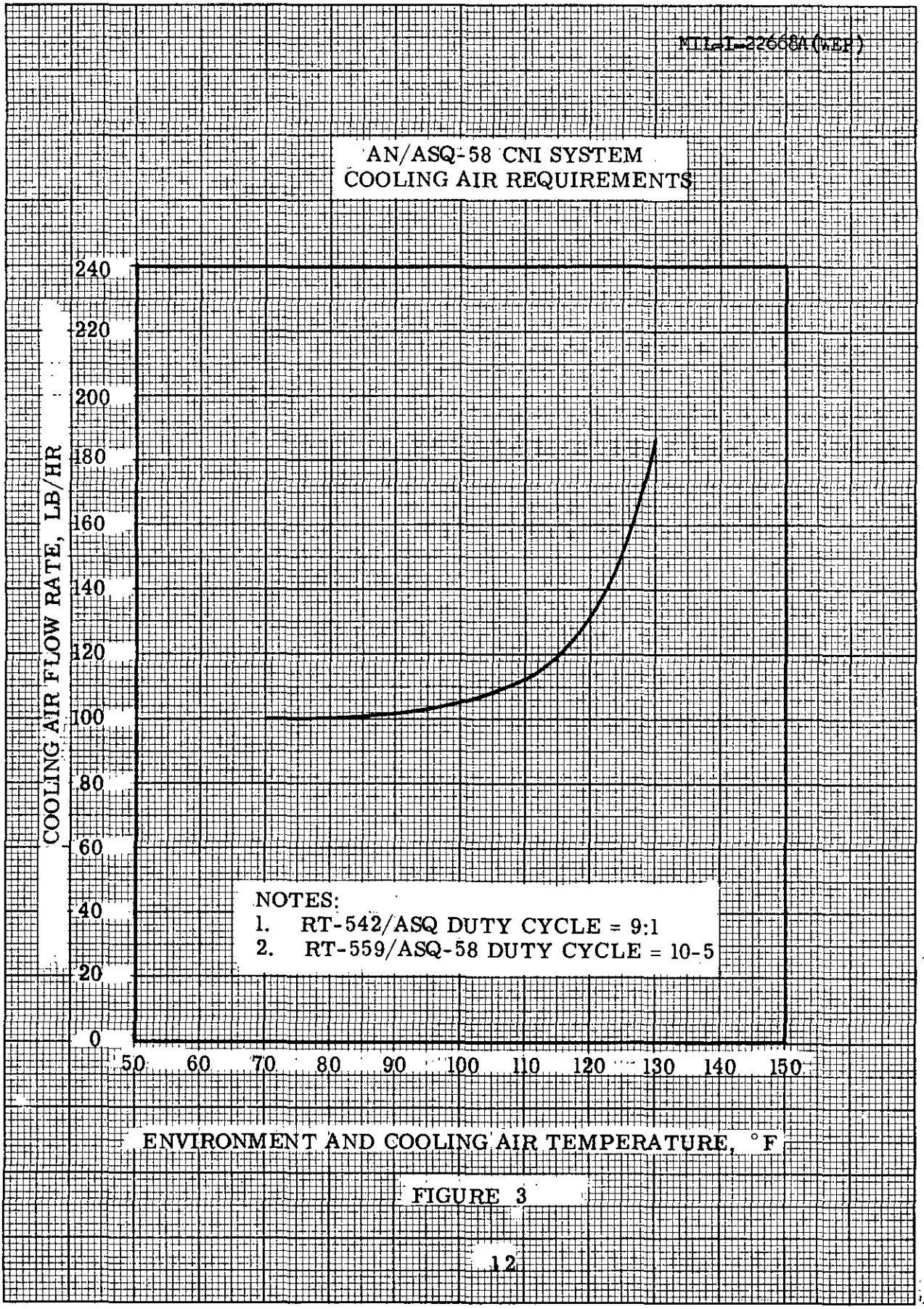


NOTE:
1. RT-542/ASQ DUTY CYCLE = 9:1

FIGURE 2

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AN/ASQ-58 CNI SYSTEM
COOLING AIR REQUIREMENTS



NOTES:
1. RT-542/ASQ DUTY CYCLE = 9:1
2. RT-559/ASQ-58 DUTY CYCLE = 10-5

FIGURE 3

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3.2.16 Undervoltage Protection - The centrals shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.2.17 Overload Protection - Overload protection for the equipments shall be in accord with MIL-E-5400. Overload protection for the centrals shall be provided in the aircraft. The use of fuses and other protective devices shall be held to a minimum in the equipments.

3.3 Detail Requirements -

3.3.1 Communication Function - The communication performance shall meet the requirements of Receiver-Transmitter, Radio RT-542/ASQ Specification MIL-R-22678(WEP).

3.3.2 Automatic Relay Communication Function - The Automatic Relay Communication function shall meet the requirements of Receiver-Transmitter, Radio RT-559/ASQ-58 Specification MIL-R-22679(WEP).

3.3.3 TACAN Function - The navigation function shall meet the requirements of the Receiver-Transmitter, Radio RT-541/ASQ and Decoder, Pulse KY-309/ASQ Specification MIL-R-22677(WEP).

3.3.4 Auxiliary Receiver, Power Supply and ADF Function - The Auxiliary Receiver, Power Supply and Automatic Direction Finder function shall meet the requirements of Amplifier-Power Supply-Receiver AM-2310/ASQ Specification MIL-A-22669(WEP).

*3.3.5 IFF/Coder Function - The IFF/Coder function shall meet the requirements of the Coder-Receiver-Transmitter KY-308(*)/ASQ Specification MIL-C-22672(WEP).

3.3.6 Communication Touch Tuning Control - The communication touch tuning control with remote channel and frequency indicator shall meet the requirements of UHF Communication Touch Tuning System MT-2141/ARC, C-2790/ARC, ID-758/ARC and SA-629 Specification MIL-C-22670(WEP).

3.3.7 Communication Control - The remote control function required for the operation of the communications sets shall meet the requirements of Controls, Radio Set C-1607/ARC-52 and C-2791/ARC of Radio Set AN/ARC-52 Specification MIL-R-18401.

3.3.8 Auxiliary Receiver Control - The remote control function required for the operation of the Auxiliary Receiver shall meet the requirements of Control, Receiver C-1457/ARR-40 of Receiving Set, Radio AN/ARR-40 Specification MIL-R-18400.

3.3.9 TACAN Control - The remote control function required for the operation of the TACAN shall meet the requirements of Control, Radio Set C-3146/ASQ Specification MIL-C-22671(WEP).

3.3.10 IFF Control - The remote control function required for the operation of the IFF shall meet the requirements of the Control, Radar Set C-1159/APX-6B of Radar Identification Set AN/APX-6B Specification MIL-R-7893.

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3.3.11 Coder Control - The remote control function required for the operation of the Coder shall meet the requirements of Control, Radar Set C-1272A/APA-89 of Coder Group AN/APA-89 Specification MIL-C-7953.

3.3.12 Mounting Plenum - The Mounting-Plenum function for the Integrated Electronic Central AN/ASQ-56 shall meet the requirements of Mounting-Plenum MT-2243/ASQ-56 Specification MIL-M-22673(WEP).

3.3.13 Mounting-Plenum - The Mounting-Plenum function for the Integrated Electronic Centrals AN/ASQ-57 and AN/ASQ-58 shall meet the requirements of Mounting-Plenum MT-2244/ASQ Specification MIL-M-22674(WEP).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests
(2) Production Sample Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor Demonstration Tests
(2) Service Approval Tests

4.2.1 Contractor Demonstration Tests - Contractor demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 Demonstration Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. Such data on parts may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor demonstration tests and when requested by the procuring activity, the equipment shall be delivered at the contractors expense to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

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4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181. Tests will be on a central basis.

4.2.4 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent at the contractors expense to a designated government laboratory for tests. This equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

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4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. Except that one (1) out of every ten (10) shall also be tested on a central basis. These tests shall be adequate do determine compliance with the requirements of material workmanship, operational adequacy and reliability. As a minimum each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run In Test

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4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

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4.4.2 Sampling Tests - Equipments and centrals selected for sampling tests shall first have passed the individual tests. Equipments and centrals shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments or Centrals Offered for Acceptance	Quantity of Equipments Selected for Sampling Tests	Quantity of Centrals Selected for Sampling Tests
First 10	1	1
Next 50	1	1
Next 75	1	One for each additional 100 or fraction thereof.
Next 100	1	
	One for each additional 200 or fraction thereof.	

4.4.2.1 Scope of Tests - As a minimum, each equipment and central selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment or central tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

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*4.4.3 Reliability Assurance Tests - Unless otherwise stipulated in the contract, equipments which have successfully passed the individual tests shall be tested for reliability assurance in accordance with Specification MIL-R-23094. (Procedure II)

*4.4.3.1 Test Level - The Test Level shall be Level III (from Table I of Specification MIL-R-23094)

*4.4.3.2 Accept-Reject Criteria - Figure 12 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria. MTBF Specified in 3.2.4.4 or referenced specifications apply.

*4.4.3.3 Length of Heat portion of Cycle - After stabilization at the high temperature limit required by the test level, the equipment shall be operated 4 hours during each cycle.

*4.4.3.4 Performance Characteristics to be Measured -

(1) UHF Communications - RT-542 and RT-559

Characteristic	Specification requirement	Acceptance requirement (if lower)
Receiver Sensitivity	Standard 5uv/10db Service 20uv/10db	20 uv/10db
Audio Capability	Standard not less than 250 mw	
	Service not less than 67.5 mw	Not less than 67.5 mw
RF Power Output	Standard not less than 16 w Service not less than 9 w	Not less than 9 w
Modulation Capability (input to clipping)	Not less than 80%	Not less than 75%
Tone Modulation Capability	70 - 100%	70 - 100%
Side Tone Level	Standard not less than 250 mw Service not less than 75 mw	Not less than 75 mw

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(2) Receiver/ADF/Power Supply - AM-2310 and AS-909

Characteristic	Specification requirement	Acceptance requirement (if lower)
Receiver Sensitivity	Standard 6uv/10db Service 18uv/10db	18uv/10db
Audio Capability	Standard not less than 250 mw Service not less than 67.5 mw	Not less than 67.5 mw
Power Supply Voltages		
430 VDC	370 to 505	360 to 515
275 VDC	241 to 302	230 to 310
250 VDC	218 to 278	210 to 290
130 VDC	114 to 142	110 to 145
140 VDC	137 to 143	135 to 145
-180 VDC	-148 to-207	-140 to-215
ADF 400 cps Inverter	Not less than 2.7 VAC	Not less than 2.7 VAC
ADF Motor Control	Not less than 75 VAC	Not less than 75 VAC
ADF RF Operational Test	Not more than 6 seconds for antenna to assume new heading 180° removed from previous heading	Not more than 12 seconds

(3) TACAN -KY-309 and RT-541

Characteristic	Specification requirement	Acceptance requirement (if lower)
Distance Accuracy and Sensitivity	±(0.2 mi +0.1%) for sig levels from -117 to -30 dbw	± (0.3 mi +0.1%)for sig levels from -114 to -30 dbw
Bearing Accuracy and Sensitivity	±2° with sig levels from -113 to -30 dbw	±2.5° with sig levels from -110 to -30 dbw

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Interrogation Coding	12 ± 0.5 usec	12 ± 0.5 usec
Identification	Not less than 50 mw audio with 30 db S/N ratio for sig levels from -117 to -30 dbw	Not less than 50 mw audio with 30 db S/N ratio for sig level from -114 to -30 dbw
Power Output	Standard not less than 30 dbw Service not less than 27 dbw	Not less than 27 dbw

(4) IFF/Coder - KY-308( )

Characteristic	Specification requirement	Acceptance requirement (if lower)
Receiver Center Frequency	1030 ± .5 mc	1030 ± 1.0mc
Normal Receiver Triggering Level	83 dbv	80 dbv
Interrogation Characteristics		
Mode 1	3±. 2 usec	3±. 2 usec
Mode 2	5±. 2 usec	5±. 2 usec
Mode 3	8±. 2 usec	8±. 2 usec
Transmitter Frequency	1090 ±3mc	1090 ± 3 mc
Transmitter Power	27 ± 3 dbw	NLT 24 dbw
SIF Reply Pulse Characteristics		
Bracket Pulses	20.30 ±0.1 usec	20.30 ±0.1 usec
Mode 1 (5 pulses at 2.9 usec intervals)	±.05 usec	±.05 usec
Mode 2 (12 pulses at 1.45 usec intervals)	±.05 usec	±.05 usec

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Mode 3 (6 pulses at 2.9 usec intervals)	±.05 usec	±.05 usec

4.4.3.5 Failure Criteria - Performance requirements below those specified herein (4.4.3.4) shall constitute one failure unless otherwise specified in the approved detailed test procedures (see 4.6). If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused other parts to fail.

4.4.3.6 Determination of Compliance - Compliance shall be determined in accordance with Figure 12 of Specification MIL-R-23094 (see paragraph 3.19 therein).

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.4.1 Special Test Schedule - Selection of equipments and centrals for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity).

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

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(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 500 hours duration and shall be conducted on equipments or centrals that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following; except that when a production sample equipment is requested and submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments or centrals which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. Voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

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4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment or central which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

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4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - The Electronic Centrals covered by this specification shall provide communication, navigation, and identification facilities required in naval aircraft to carry out military missions.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC, 26 ±1.0 V AC and 115 ±1.0 V AC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

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6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

(4) Reordered Equipment - The following specifications shall take precedence over those concerning performance requirements listed or referenced in paragraph 2.1 where equipment is reordered except in those areas where tne contract specifies redesign or modification relating to performance.

MIL-E-5400E
MIL-I-6181B

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6.6 An asterisk "*" preceding a paragraph number denotes a change made in that paragraph from the previous version of the document. (See paragraph 4 of the Explanation page i.)

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1).

6.8 The parentheses (*), when used in the type designation, may be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.10 The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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